

Prateek Chandra Jha

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SKILLS

PROGRAMMING

Python, Jupyter/iPython Notebook
SQL, NumPy, Pandas, Dask, PyMC,
Huggingface Transformers
FastAPI, Streamlit, LangChain
JAX, TensorFlow, PyTorch, Scikit-Learn
DeepSpeed, RAPIDS (cuML),
GPU, Docker, AWS Sagemaker,
Git, $\LaTeX$

MACHINE LEARNING

Regression, Gradient Boosting
(XGBoost, LightGBM, CatBoost),
Deep Learning (CNNs, RNNs,
Transformers, GANs),
Large Language Models (LLMs),
Reinforcement Learning,
Clustering (K-Means, DBSCAN,
HDBSCAN), Transfer Learning,
Anomaly Detection,
Graph Neural Networks (GNN),
Federated Learning, AutoML,
Time Series Analysis,
Explainable AI (XAI),
Causal Inference

EDUCATION

MSC. DATA SCIENCES

CHENNAI MATHEMATICAL
INSTITUTE

May 2021 | Chennai, India

MSC. FINANCIAL ENGINEERING

WORLDQUANT UNIVERSITY
February 2021 | Online

B.TECH, AEROSPACE ENGG.

IIT BOMBAY
August 2015 | Mumbai, India

LINKS

Github:// [prateekchandrajha](#)
LinkedIn://
[prateek-chandra-j-9906632a](#)
Twitter:// [@prateekjha](#)
Instagram:// [prateekcjha](#)

APPLIED ML EXPERIENCE

AMAZON | APPLIED SCIENTIST

Jul 2021 – Present | Bangalore (India)

- **Amazon Gift Cards Compliance, Abuse & Anti-Money Laundering ML Models**
Developed ML models which prioritized (ranked) manual investigations by Suspicious Activities Monitoring teams predicting high-risk accounts. Models demonstrated increase in true positive rate from an average of 23% pre-deployment to 85% post-deployment, enhancing investigation efficiency. Investigators have revoked >\$2MM from suspicious accounts.
- **Branded Gift Cards (BGC) Category Recommender ML Models**
Developed a 2-stage ML recommender model to enhance customer targeting for BGC brand categories. A/B tests demonstrated 70.5% (US) and 106% (UK) lift on baseline CTRs, resulting in predicted annualized gains of \$77.1MM for the US and £8.5MM for UK. Designed incrementality A/B experiments for new customer acquisitions via Amazon Display Ads, optimizing for expense as % of NPV (Net Present Value) gains.
- **Generative AI-Powered Personalized Gift Card Designs**
Leveraged Amazon Bedrock's Titan Image and Text Generation models to enable customers to create personalized gift card designs. These enhancements incorporated advanced visual customization (inpainting, outpainting, generative re-sizing for multiple variants), intelligent occasion suggestions, and context-aware gifting message generation.
- **Physical Gift Cards Sales Forecasting Resolving Supply Chain Bottlenecks**
Developed sales forecasting models to improve demand accuracy for physical gift cards with a 6-9 month vendor lead time. This led to a recommended 20% reduction in inventory holding periods, projected to enhance economic profit by \$5.1MM and improve free cash flow by \$4.4MM due to decreased holding costs.

RESEARCH EXPERIENCE

Social Network Topology of Gift Card Transactions (AMLC 2023^a)

- By segmenting connected components of gift cards transaction networks using directed graph isomorphism, we identify how to drive transitions between the 13 directed graph isomorphism classes for networks of size 3 to enhance gifting reciprocity.

Quantifying Network Effects for A/B Experiments (CSS 2024^b)

- Traditional A/B testing often misses network effects among interconnected users. In our annual review of over 60 global gift card experiments, we found treatment group gains often fall short of Amazon's profitability metrics. To address this, we quantified network effects to refine our estimations. We use cluster-based experimental designs to better understand how network behaviors influence outcomes.

Geospatial Optimization of Marketing Campaigns (AGC 2023^c)

- Developed a ZIP+4 personalized geofencing strategy for Amazon gift cards, enabling hyper-localized, targeted campaigns. By leveraging ZIP+4 codes, we pinpoint specific neighborhoods, streets, or buildings to create tailored marketing messages. For example, in Manhattan, ZIP+4 can target an office building on West 26th Street for tech-related gift cards, aligning with the professional demographics and purchasing power of the area.

^aAmazon Machine Learning Conference

^bConsumer Science Summit

^cAmazon Geospatial Conference